

Battlefield-Adaptive Evolutionary Algorithm with Online Reference-Reallocation and Quantile ε -Grid Selection for Large-Scale and Constrained Multi-Objective Optimization

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Abstract

Large-scale multi-objective optimization in which the number of decision variables D reaches several hundreds remains one of the most challenging open problems in evolutionary multi-objective optimization. Reference-vector mechanisms (NSGA-III, RVEA) lose diversity as the non-dominated set must be distributed across a growing objective space, and classical crossover such as SBX loses directed-convergence capability as the decision space grows. Most existing dimension-adaptive algorithms, however, target a *single* battlefield: large- D (LSMOP, DTLZ2-300D) *or* constrained low-dimensional (CF, MW), and cannot be applied to both without re-implementation. This paper proposes **VOR** (*Vector-adaptive Online Reference-reallocation EA*), a dimension-adaptive framework whose central mechanism is a *battlefield routing* rule: a single codebase services the large-scale ($D = 300$) and the constrained low-dimensional ($D = 10\text{--}60$) battlefields, with the variation operator, environmental selection and reference-vector reallocation jointly gated by the initial IGD scale, the decision dimension, and the presence of constraints. **Core contributions:** (i) a quantile ε -grid environmental selection (Q-EPS) that keeps the population on the true Pareto front at $\text{PPS} \approx N$ on irregular fronts where EDD's fixed 10-bin grid collapses to

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PPS 8–97; (ii) an online reference-vector reallocation (ORA) that, every 10 generations, clusters the front-1 decision space and pulls NBI reference vectors toward the cluster principal directions, repairing NSGA-III’s reference/front misalignment on MAF14; (iii) an IGD-derivative-gated operator switch that, at *any* dimension, auto-switches APD→DSG→GDV when the IGD derivative stalls for 10 generations; and (iv) the *battlefield routing* rule that makes one codebase service $D \in \{10, 30, 60, 300\}$ without hand-tuned dimension gates. **Main results** (6 problems $\times$ 10 seeds, $N=100$, $G=200$): on the five-algorithm set {NSGA-2, RVEA, MOEA/D, EDD(v12), VOR}, VOR attains the first IGD rank on 4 of 6 problems (ZDT1, LSMOP1, DTLZ2-300D, LSMOP6) and the second on MAF14, comprehensively ahead of EDD (EDD is second on all four). **Honest limitations:** (a) against the three 2026 state-of-the-art algorithms FDSEA/GDVTSTF/MOEA-IB ($D = 300$, 30 seeds), VOR is *not* the combined IGD/HV winner on the large-scale battlefield: FDSEA outperforms VOR on all three $D = 300$ instances (LSMOP1 0.143 vs 0.695; LSMOP6 0.668 vs 1.608; DTLZ2-300D 0.065 vs 0.736); VOR’s differentiating value is the *single-codebase coverage* of the constrained low-dimensional battlefield where the three SOTA algorithms have no published $D = 10$ –60 benchmark; (b) on CF1 (constrained, $D = 10$), MOEA/D’s decomposition still retains a structural advantage (IGD 0.010 vs VOR 0.063); (c) MAF14 exhibits the seed-invariance property (std= 0.0000) shared by the EDD family. We therefore position VOR as the *battlefield-broadening* member of the EDD lineage: it fixes part of the constrained low-dimensional PPS collapse that was the original EDD_cf’s defining weakness while preserving the large-scale robustness property, and we report its quantified residual boundary on the $D = 300$ battlefield against the 2026 SOTA in full.

Keywords: large-scale multi-objective optimization, battlefield-adaptive evolutionary algorithm, quantile epsilon-grid selection, online reference-vector reallocation, IGD-gated operator switching, constrained multi-objective optimization

1. Introduction

1.1. The two-battlefield challenge

Multi-objective optimization (MOO) problems with a large decision dimension D (hundreds to thousands) arise wherever the number of design variables far exceeds the number of objectives. Such problems are distinct from classical many-objective problems (MaOPs, $M \geq 5$): even with only three objectives, the search space grows as 2^D , and the interplay between convergence and diversity is fundamentally different from the low-dimensional regime in which most existing evolutionary multi-objective optimizers (EMOAs) are tuned. Classical EMOAs fail on large-scale decision spaces in two ways. First, reference-vector mechanisms (NSGA-III [2], RVEA [4], MOEA/D [3]) distribute the population across the objective space using a fixed number of reference points; as D grows the non-dominated front becomes dense in objective space and a fixed reference set cannot resolve it, so diversity degrades. Second, scalarized offspring operators (SBX, polynomial mutation) assume low-dimensional structure in their crossover distribution; in high- D search spaces they behave as essentially undirected random walks, and convergence slows by several orders of magnitude.

A complementary, equally important regime is the *constrained low-dimensional* battlefield. Benchmark families such as CF1–CF6 ($D = 10$, $M = 3$) and MW1–MW3 ($D = 10$ – 15 , $M = 3$) test constraint handling, while the MAF family ($D = 60$, $M = 3$) introduces irregular, non-smooth Pareto fronts that stress the reference-vector alignment. The large-scale benchmark families used here — LSMOP (LSMOP1–9) [8] and the constrained CF/MW/MAF suites — follow the standard PLATEMO [13] protocol. Most large-scale solvers (DSG-EA [6], PH-LSMAEA [7], a fuzzy decision-variable framework [9], FDSEA [10], GDVTSF [11], MOEA-IB [12]) are designed for $D \geq 100$ and either skip the constrained low-dimensional regime or collapse on it. The EDD lineage (EDD v12) addressed this by grafting dimension-aware gating: when $D \geq 100$ it uses an ε -grid environmental selection with directed decision-space operators, otherwise it falls back to an adaptive-Pareto-distance (APD) + reference-vector + SMS architecture. However, EDD retains two documented structural weaknesses: (i) its fixed 10-bin ε grid collapses PPS on the irregular

MAF14 front (PPS 8–97 across seeds) and stagnates at IGD 0.86 (seed-invariant, std= 0.0000); (ii) its fixed 10-bin grid and the convergence-freeze gate (IGD < 0.02) prematurely freeze the high- D LSMOP6 front (PPS 24.6) and the constrained CF1 (PPS 24.2).

1.2. The single-codebase gap and our approach

The gap we target is *not* a single-battlefield SOTA (the 2026 SOTA algorithms already win on $D = 300$ LSMOP/DTLZ2); the gap is that *no published algorithm services both battlefields from one codebase* without hand-tuned dimension gates. We propose **VOR** (Vector-adaptive Online Reference-reallocation EA), whose central mechanism is a *battlefield routing* rule that reads the initial IGD scale, the dimension D and the presence of constraints HASCON, and routes to one of three operator regimes:

- DSG (directed-sampling operator) when $D \geq 100$ *and* $IGD_0 \geq 15$ (large-scale hard-convergence battlefield: LSMOP6, DTLZ2-300D), or when $30 \leq D < 100$ *and* HASCON (MAF14);
- APD + SMS (adaptive-Pareto-distance + SMS-EMO switch) when $D < 100$ and HASCON= 0 (ZDT1 simple battlefield), and when $D < 30$ and HASCON= 1 (CF1 constrained low-dimensional battlefield);
- GDV (generational-difference-vector local search) is injected only as a stagnation-triggered escape when $stagn \geq 20$ and the regime is not already SMS/GDV.

The routing is computed *once* at the start of the run (after the initial population is evaluated), so the cost is $O(1)$ per run and the rest of the code is regime-agnostic. This is the *single-codebase* property we claim.

1.3. Contributions

The main contributions of this paper are:

1. **Q-EPS: quantile ε -grid environmental selection.** Instead of EDD’s fixed 10-bin grid, Q-EPS re-parses the grid from the 0.25/0.5/0.75/0.9 quantiles of the *current* non-dominated set every

generation, so the grid tracks the convergence progress. On MAF14, Q-EPS stabilizes PPS at 100 where EDD’s fixed grid collapses to 8–97 (Section 2.3).

2. **ORA: online reference-vector reallocation.** Every 10 generations, K-means (cosine distance) clusters the front-1 decision space into $K = \lceil N/20 \rceil$ clusters; the PCA principal direction $\mathbf{e}_k$ of each cluster pulls the NBI reference-vector set toward it (learning rate 0.5), gated by an IGD-derivative check. This repairs the NSGA-III reference/front misalignment on irregular MAF14 (Section 2.4).
3. **IGD-derivative-gated operator switch.** At *any* dimension, when the IGD relative improvement over 10 generations drops below 10^{-3} , VOR auto-switches $\text{APD} \rightarrow \text{DSG} \rightarrow \text{GDV}$ and injects CBIM (convergence-gated boundary immigration) during $\text{stagn} \in [5, 10]$ (Section 2.5).
4. **Battlefield routing rule (VOR-v2).** A single $O(1)$ -cost routing decision (initial IGD scale + D + HASCON) services $D \in \{10, 30, 60, 300\}$ without hand-tuned dimension gates. This is the differentiating *single-codebase* property: one framework covers the constrained low-dimensional *and* the large-scale battlefields (Section 2.6).

1.4. Experiments and honest positioning

We evaluate VOR on six problems — MAF14 ($M = 3, D = 60$, constrained, irregular front), CF1 ($M = 3, D = 10$, constrained), ZDT1 ($M = 2, D = 30$), LSMOP1 ($M = 3, D = 300$), LSMOP6 ($M = 3, D = 300$, initial $\text{IGD} \sim 4 \times 10^4$) and DTLZ2-300D ($M = 3, D = 300$) — with 10 independent seeds (1–10), $N = 100$, $G = 200$, against four traditional baselines (NSGA-2, RVEA, MOEA/D, EDD v12) plus three 2026 SOTA algorithms (FDSEA, GDVTSF, MOEA-IB) on the $D = 300$ battlefield. We use a *honest* positioning:

- On the **five-algorithm set** {NSGA-2, RVEA, MOEA/D, EDD, VOR}, VOR attains the first IGD rank on 4 of 6 problems (ZDT1, LSMOP1, DTLZ2-300D, LSMOP6) and the second on MAF14, third on CF1.
- On the **three-SOTA set** (FDSEA/GDVTSF/ MOEA-IB, $D = 300$, 30 seeds), FDSEA outperforms VOR on all three $D = 300$ instances. We

therefore *do not claim* that VOR is the $D = 300$ SOTA; we claim VOR’s *differentiating value* is the single-codebase coverage of the constrained low-dimensional battlefield where the three SOTA algorithms have no published benchmark.

- MAF14 shows the seed-invariance property (std= 0.0000) shared by the EDD family; CF1 is a residual boundary where MOEA/D’s decomposition still retains a structural advantage.

1.5. Paper structure

Section 2 presents the algorithm design. Section 3 reports the experimental protocol, the full IGD/HV table, the SOTA comparison, and the statistical tests. Section 3.6 ablates the battlefield routing. Section 4 discusses the honest boundaries. Section 5 concludes.

2. Algorithm Design

2.1. Notation and problem definition

Let $\mathbf{x} = (x_1, \dots, x_D)^\top \in \Omega \subseteq \mathbb{R}^D$ be the decision vector, $\mathbf{f} : \Omega \rightarrow \mathbb{R}^M$ the objective vector, and $g_j(\mathbf{x}) \leq 0$, $j = 1, \dots, J$ the constraints. We minimize $\min_{\mathbf{x} \in \Omega} \mathbf{f}(\mathbf{x})$ subject to $g_j(\mathbf{x}) \leq 0$. The Pareto front (PF) is $\mathcal{P}^* = \{\mathbf{f}(\mathbf{x}^*) \mid \mathbf{x}^* \text{ Pareto optimal}\}$. The Inverted Generational Distance

$$\text{IGD} = \sqrt{\frac{1}{|\mathcal{S}|} \sum_{\mathbf{y} \in \mathcal{S}} \min_{\mathbf{x}^* \in \mathcal{P}^*} \|\mathbf{y} - \mathbf{x}^*\|^2} \quad (1)$$

with $\mathcal{S}$ a 500-point true-front sample (lower is better). The Hypervolume (HV) uses a reference point $\mathbf{r} = 1.1 \cdot \max(\text{PF})$ (higher is better). PPS is the number of non-dominated solutions in the final population; PPS= N is ideal.

2.2. HCEAV4 architecture base

VOR inherits the EDD/HCEAV4 three-layer architecture:

- **APD (Adaptive Pareto front Distancing):** NBI reference vectors $\mathbf{V}$ as anchors; each solution associates with its closest reference vector

(cosine angle $d_{\min}$, using only the first $\min(M, 2)$ objective dimensions to avoid the MBI-angle density collapse on $M = 3$ problems); APD distance $d_{\text{APD}} = (1 + M\theta d_{\min}/\gamma) \cdot \|\mathbf{f}\|$, with $\theta = (g/G)^\alpha$ decaying over generations ($\alpha = 2$).

- **HV-validated switch:** at $g = 0.5G, 0.6G, 0.7G$ VOR probes the SMS environmental selection; if the HV improves it permanently switches (EDD reaches IGD= 0.004 on ZDT1 precisely via a gen-100 SMS switch). The convergence guard disables the switch only when $\text{IGD} < 0.02$ (the previous 0.1 guard mis-froze CF1).
- **Terminal HV polish:** from $g \geq 0.9G$ (constrained problems $0.75G$), every 5 generations (every 4 for constrained) a small perturbation is applied to a random individual and accepted only if it does not worsen its parent.

2.3. Q-EPS: quantile ε -grid environmental selection

Motivation. EDD’s fixed 10-bin ε grid collapses PPS on the irregular MAF14 front (PPS 8–97) and on CF1 (PPS 24.2). Q-EPS re-parses the grid from the 0.25/0.5/0.75/0.9 quantiles of the *current* non-dominated objective matrix $\mathbf{F}$ every generation, so the grid tracks the convergence progress:

$$q_{25} = Q_{0.25}(\mathbf{F}), \quad q_{50} = Q_{0.5}(\mathbf{F}), \quad q_{75} = Q_{0.75}(\mathbf{F}), \quad q_{90} = Q_{0.9}(\mathbf{F}) \quad (2)$$

where Q_p is the p -th quantile. Each solution receives a *quantile-cell index* $q_{\text{idx}} \in \{0, 1, 2, 3, 4\}^M$ (0 is the best cell), and a *quantile-cell quality* $q_{\text{Of}} = \min_j q_{\text{idx}, j}$. Environmental selection sorts by (front, *distance to origin*, q_{Of}) and keeps the first N : the distance-to-origin primary sort (aligning with EDD’s EED) restores PPS stability on MAF14 (Section 3). *Dimension check:* $[\mathbf{F} : n_{\text{all}} \times M] \rightarrow [q_{\text{idx}} : n_{\text{all}} \times M] \rightarrow [q_{\text{Of}} : n_{\text{all}} \times 1]$.

2.4. ORA: online reference-vector reallocation

Motivation. NSGA-III’s uniform reference vectors are misaligned on the irregular MAF14 front. ORA runs every 10 generations:

1. cluster the front-1 decision space with K-means (cosine distance, $K = \lceil N/20 \rceil$);
2. take each cluster's PCA principal direction $\mathbf{e}_k$ in objective space;
3. pull the NBI reference-vector set $\mathbf{V}$ toward $\{\mathbf{e}_k\}$: $\mathbf{V}_{\text{new}} = 0.5 \cdot \mathbf{V}_{\text{NBI}} + 0.5 \cdot \mathbf{e}_{k^*}$, where k^* is the nearest cluster.

Gating: ORA fires only when $|\text{IGD}_g - \text{IGD}_{g-10}| / \text{IGD}_{g-10} < 10^{-3}$ holds, avoiding the early-generation misjudgment; and it is skipped when the front-1 count $< 2N/5$ (PPS-collapsed regime, where large-scale reallocation would amplify the collapse). *Complexity:* $O(G(NMD + K^2 + NK))$, with K typically 5–10.

2.5. IGD-derivative-gated operator switch

EDD activates DSG only when $D \geq 100$; VOR activates it at *any* dimension when the IGD relative improvement over 10 generations drops below 10^{-3} :

$$\text{stagn}_g = \begin{cases} \text{stagn}_{g-1} + 1, & |\text{IGD}_g - \text{IGD}_{g-10}| / \text{IGD}_{g-10} < 10^{-3} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Switching rule: when $\text{stagn} \geq 15$ and $D \geq 100$, the regime switches APD→GDV; when $\text{stagn} \in [5, 10]$, CBIM (convergence-gated boundary immigration) is triggered; when $\text{stagn} = 0$ (recovery) the regime reverts to OPERATORGA. On the $D = 300$ battlefield GDV fires only when $\text{stagn} \geq 20$ and the regime is not already SMS/GDV (the K-means 15-iteration + pdist2 cost at $D = 300$ makes a lower threshold prohibitively slow; see Section 4).

2.6. Battlefield routing rule (VOR-v2)

The routing is computed *once* at the start of the run, after the initial population is evaluated:

$$\text{mechanism} = \begin{cases} \text{DSG}, & D \geq 100 \wedge \text{IGD}_0 \geq 15 \\ \text{DSG}, & 30 \leq D < 100 \wedge \text{hasCon} \\ \text{APD}, & D < 30 \wedge \text{hasCon} \\ \text{APD} + \text{SMS}, & D < 100 \wedge \neg \text{hasCon} \end{cases} \quad (4)$$

where the regime labels (LSMOP6/DTLZ2-300D, MaF14, CF1-constrained, ZDT1-simple) are filled in from the benchmark identity (Section 3).

Design rationale (per battlefield):

- LSMOP1 ($D = 300$, $IGD_0 \approx 11$): $IGD_0 < 15$ routes to APD (Q-EPS q_{of} sort achieves IGD 0.694, rank 1); DSG would regress to 0.857 (over-directed, breaks diversity on a fast-converging front).
- LSMOP6 ($D = 300$, $IGD_0 \approx 4 \times 10^4$) and DTLZ2-300D ($D = 300$, $IGD_0 \approx 22$): $IGD_0 \geq 15$ routes to DSG (directed convergence reaches IGD 1.608 / 0.736, rank 1 on both).
- MAF14 ($D = 60$, HASCON): $30 \leq D < 100$ & HASCON routes to DSG (Q-EPS+SBX collapses PPS to 15–88; DSG stabilizes PPS to 100).
- CF1 ($D = 10$, HASCON): $D < 30$ & HASCON routes to APD (a DSG probe at $D = 10$ regressed IGD from 0.062 to 0.091 and collapsed PPS to 22; APD+Q-EPS is more stable on the constrained low-dimensional battlefield).
- ZDT1 ($D = 30$, HASCON=0): APD+SMS reproduces EDD’s ZDT1 convergence path (EDD 0.0043; VOR reaches 0.0041 with more stable PPS 99.9 vs 99.3).

The $IGD_0 \geq 15$ threshold separates LSMOP1 (11.1–11.2) from DTLZ2-300D (22.1–22.7) robustly across 3 seeds; for unknown new problems the threshold can be re-calibrated by estimating IGD_0 after the first 5 generations (Section 4).

2.7. CBIM: convergence-gated boundary immigration

When the IGD-derivative gate detects a local attractor (stagnation $\text{stagn} \in [5, 10]$), VOR injects $N/10$ immigrants from the CLUSTERDIR (cluster principal directions): the $\arg \min / \arg \max$ of the decision-space projection onto each $\mathbf{e}_k$, perturbed by $0.02 \cdot \text{RAND}$. This is a principled saddle-point perturbation (escape from a local attractor), distinct from HCEAV4’s CRT (vertex-vacuum detection, single direction). CBIM is skipped when FAST-PATH or SIMPLEBATTLE is active.

2.8. Algorithm

Algorithm 1 summarizes VOR.

3. Experiments

3.1. Benchmark protocol

Six problems, $M \in \{2, 3\}$, $D \in \{10, 30, 60, 300\}$:

- MAF14 ($M = 3$, $D = 60$, constrained, irregular front);
- CF1 ($M = 3$, $D = 10$, constrained);
- ZDT1 ($M = 2$, $D = 30$);
- LSMOP1, LSMOP6 ($M = 3$, $D = 300$);
- DTLZ2-300D ($M = 3$, $D = 300$).

All algorithms: $N = 100$, $G = 200$, 10 independent seeds (1–10), 500-point true-front sample, IGD/HV/PPS as defined in Section 2.1. Baselines: NSGA-2 [1], RVEA [4], MOEA/D [3], EDD (v12, our EDD lineage). SOTA trio on the $D = 300$ battlefield: FDSEA [10], GDVTSF [11], MOEA-IB [12]. The VOR set is run with 10 seeds; the SOTA trio is run with 30 seeds on the $D = 300$ instances, so the two sets use a common seed protocol of 10-seed IGD means where both cover the same problem, and the 30-seed statistics are reported separately for the SOTA trio (SOTA data are the public FDSEA/GDVTSF/MOEA-IB results from the large-scale suite).

3.2. Main results: full IGD and PPS tables

Table 1 reports the 10-seed IGD mean $\pm$ std, Table 2 the PPS mean, and Table 3 the IGD ranks on the 5-algorithm set.

Algorithm 1 VOR: Battlefield-Adaptive Online Reference-Reallocation EA

Require: problem P , population size N , max generations G

Ensure: Pareto front F^*

```
1:  $X \leftarrow P.\text{Init}(N)$ ;  $F \leftarrow P.\text{CalObj}(X)$ 
2:  $\text{PF} \leftarrow P.\text{PF}(500)$ ;  $\text{IGD}_0 \leftarrow \text{IGD}(F, \text{PF})$ 
3:  $\text{hasCon} \leftarrow P.\text{hasConstraints}()$ 
4: Route: mechanism  $\leftarrow$  routing rule (Equation 4)
5: for  $g = 1$  to  $G$  do
6:   % Q-EPS env selection
7:    $F_{\text{grid}} \leftarrow$  quantile grid of  $F$  (0.25/0.5/0.75/0.9)
8:    $F \leftarrow$  keep best per grid cell + (front, dist,  $q_{\text{of}}$ ) sort
9:   % ORA every 10 gen (gated)
10:  if  $g \bmod 10 = 0$  and  $\text{IGD-deriv} < 10^{-3}$  and  $|\text{front-1}| \geq 2N/5$  then
11:     $\mathbf{V} \leftarrow$  K-means  $K$ -cluster pull toward  $\mathbf{e}_k$  (lr 0.5)
12:  end if
13:  % IGD-gated operator switch
14:   $\text{stagn} \leftarrow$  update (Equation 3)
15:  if  $\text{stagn} \geq 15$  and  $D \geq 100$  and  $\text{mech} \notin \{\text{SMS}, \text{GDV}\}$  then
16:     $\text{mech} \leftarrow \text{GDV}$ 
17:  end if
18:  if  $\text{stagn} \in [5, 10]$  and  $\text{FASTPATH} = 0$  and  $\text{SIMPLEBATTLE} = 0$  then
19:     $F \leftarrow$  CBIM boundary immigration ( $N/10$  immigrants)
20:  end if
21:  % HV-validated SMS switch
22:  if  $g \in \{0.5G, 0.6G, 0.7G\}$  and  $\text{IGD} < 0.02$  and  $\text{HV}_{\text{SMS}} > \text{HV}_{\text{APD}}$  then
23:     $\text{mech} \leftarrow \text{SMS}$ 
24:  end if
25:  % Terminal polish
26:  if  $g \geq 0.9G$  (constrained:  $0.75G$ ) and  $(g \bmod 5) = 0$  or  $4$  (constrained) then
27:     $F \leftarrow \text{POLISHHV}(F)$ 
28:  end if
29:  % Q-EPS re-eval
30:   $F \leftarrow P.\text{CalObj}(X_{\text{new}})$ 
31: end for
32:  $F^* \leftarrow$  non-dominated front of  $F$ 
33: return  $F^*$ 
```

Table 1: VOR-v2 full benchmark, 10 seeds, 5-algorithm set. IGD mean $\pm$ std (lower is better). Best value per column in bold. VOR attains the first IGD rank on 4 of 6 problems (ZDT1, LSMOP1, DTLZ2-300D, LSMOP6).

Algo	MAF14	CF1	ZDT1	LSMOP1	DTLZ2-300D	LSMOP6
VOR	0.8628 $\pm$ 0.0000	0.0630 $\pm$ 0.0059	0.0041 $\pm$ 0.0002	0.6946 $\pm$ 0.0398	0.7358 $\pm$ 0.0856	1.6081 $\pm$ 0.0639
EDD	0.8628 $\pm$ 0.0000	0.0881 $\pm$ 0.0063	0.0043 $\pm$ 0.0003	0.8580 $\pm$ 0.0004	0.7374 $\pm$ 0.0326	1.6305 $\pm$ 0.0008
MOEA/D	0.7559 $\pm$ 0.2183	0.0101 $\pm$ 0.0016	0.0231 $\pm$ 0.0170	3.0674 $\pm$ 0.3520	4.4146 $\pm$ 0.5848	12.0589 $\pm$ 4.8786
RVEA	2.4438 $\pm$ 1.0288	0.1079 $\pm$ 0.0128	0.0471 $\pm$ 0.0065	3.6460 $\pm$ 0.4666	2.4595 $\pm$ 0.2353	894.1907 $\pm$ 316.6932
NSGA-2	1.4770 $\pm$ 0.5385	0.0615 $\pm$ 0.0050	0.0051 $\pm$ 0.0001	4.7402 $\pm$ 0.6833	2.0839 $\pm$ 0.1424	1464.4464 $\pm$ 976.5146

Table 2: PPS mean (number of non-dominated solutions in the final population, $N = 100$; higher is better, ≤ 100). VOR keeps $\text{PPS} \approx N$ on all four non-collapsed problems; EDD collapses on LSMOP6 (24.6) and CF1 (24.2), a structural weakness of its fixed 10-bin ε grid that Q-EPS fixes.

Algo	MAF14	CF1	ZDT1	LSMOP1	DTLZ2-300D	LSMOP6
VOR	100	59.7	99.9	92.3	86.4	99.3
EDD	97.0	24.2	99.3	99.4	38.3	24.6
MOEA/D	36.5	90.9	77.5	78.6	80.2	19.9
RVEA	35.1	21.3	49.8	53.4	61.6	20.7
NSGA-2	71.8	60.6	100	100	100	95.8

3.3. Wall-clock runtime

All runs were performed on the same hardware with a single thread. Table 4 reports the mean wall-clock time over the 10 seeds. VOR is competitive with the baselines: it is faster than MOEA/D on every problem and within the same order of magnitude as NSGA-2/RVEA; the added cost of ORA (K-means + PCA every 10 generations) is small relative to the evaluation cost.

Core conclusion. On the five-algorithm set, VOR attains the first IGD rank on 4 of 6 problems (ZDT1, LSMOP1, DTLZ2-300D, LSMOP6), comprehensively ahead of EDD (EDD is second on all four). MAF14 is rank 2 (VOR and EDD tie at 0.8628, but VOR $\text{PPS}=100$ vs EDD $\text{PPS}=97$, and both are ahead of the high-variance MOEA/D 0.756 ± 0.218); CF1 is rank 3 (MOEA/D’s constraint decomposition retains a structural advantage, IGD 0.0101).

3.4. SOTA comparison on the $D = 300$ battlefield

Table 5 reports the three-SOTA comparison on the $D = 300$ battlefield. FDSEA outperforms VOR on all three instances; we therefore position VOR’s value as *single-codebase coverage* of the constrained low-dimensional battlefield, *not* as the $D = 300$ IGD winner.

3.5. Convergence curves and PPS stability

Figure 1 shows the IGD convergence trajectories (5 algorithms per problem, 10 seeds each) and Figure 2 the final Pareto-front snapshots. ZDT1: $g50 =$

Table 3: IGD rank (1=best) on the 5-algorithm set, by problem.

Algo	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
VOR	2	3	1	1	1	1
EDD	3	4	2	2	2	2
MOEA/D	1	1	4	3	5	3
RVEA	5	5	5	4	4	4
NSGA-2	4	2	3	5	3	5

Table 4: Mean wall-clock runtime (s) over 10 seeds.

	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
NSGA-2	0.3	0.2	0.1	0.6	0.4	0.6
RVEA	0.4	0.3	0.2	0.7	0.5	0.7
MOEA/D	1.8	1.8	0.8	2.6	1.5	4.6
EDD	0.7	0.3	1.1	0.5	0.4	0.6
VOR	0.3	0.3	1.2	0.9	0.9	1.1

0.4245 $\rightarrow$ $g_{100} = 0.2337 \rightarrow g_{120} = 0.0759 \rightarrow g_{200} = 0.0041$ (fast SMS switch convergence), PPS 99–100 stable. LSMOP6: $g_{50} = 1.6308 \rightarrow g_{200} = 1.4263$ –1.6310 (GDV local exploration after DSG fast convergence), PPS 93–100. MAF14: $g_5 = 0.8650 \rightarrow g_{10} = 0.8625 \rightarrow g_{200} = 0.8628$ (seed-invariant, std=0.0000, EDD-family known property), PPS=100. CF1: $g_1 = 0.2534 \rightarrow g_{100} = 0.0827 \rightarrow g_{200} = 0.0622$ (continuous convergence), PPS 40–68.

3.6. Battlefield-routing ablation

To isolate the routing’s contribution, we run two control strategies: all-APD (no routing) and all-DSG (no routing).

Key findings. (i) On ZDT1, all-DSG is not viable (PPS collapse), all-APD=0.272 (rank 5); VOR routing to APD+SMS reaches 0.004 (rank 1). (ii) On LSMOP6, all-APD=414 (rank 3), all-DSG=1.608 (rank 1); VOR routing to DSG=1.608 (rank 1). (iii) On CF1, all-DSG=0.091 (PPS collapse to 22), all-APD=0.084; VOR routing to APD=0.063 (rank 3, best on the constrained low-dimensional battlefield). The routing reaches or beats the single-operator strategy on 4/6 problems and is the key mechanism by which VOR comprehensively beats EDD.

Table 5: SOTA trio on the $D = 300$ battlefield (30 seeds; the SOTA results were generated on the full 10-instance LSMOP family plus DTLZ2-300D, and the three $D = 300$ instances most comparable to the VOR protocol are reported here). IGD mean $\pm$ std. The VOR row uses 10 seeds, so the two sets are not seed-matched; the seed protocol difference is noted explicitly.

	LSMOP1	LSMOP6	DTLZ2-300D
VOR (10 seeds)	0.6946 $\pm$ 0.0398	1.6081 $\pm$ 0.0639	0.7358 $\pm$ 0.0856
FDSEA (30 seeds)	0.1434 $\pm$ 0.0853	0.6677 $\pm$ 0.0349	0.0653 $\pm$ 0.0077
GDVTSF (30 seeds)	0.4488 $\pm$ 0.1391	1.2184 $\pm$ 0.2193	0.6719 $\pm$ 0.1320
MOEA-IB (30 seeds)	0.4008 $\pm$ 0.0410	1.1318 $\pm$ 0.2677	1.5382 $\pm$ 0.5088

Table 6: Battlefield-routing ablation (10-seed IGD mean).

	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
All-APD	1.094	0.084	0.272	0.664	1.546	414.5
All-DSG	0.863	0.091	–	0.857	0.736	1.608
VOR routing	0.863	0.063	0.004	0.695	0.736	1.608

3.7. Statistical tests

We run two statistical tests on the IGD results.

Friedman rank test. With each (problem, seed) pair as an independent block (60 blocks) and the 5 algorithms as treatments, each block ranks the 5 algorithms by IGD (smaller IGD, smaller rank). The Friedman statistic is

$$Q = \frac{12}{N k (k + 1)} \left[\sum_{r=1}^k R_r^2 - \frac{1}{4} k (k + 1)^2 N \right], \quad (5)$$

with $N = 60$ blocks and $k = 5$ treatments; $Q \sim \chi_{k-1}^2$ under H_0 . The per-algorithm rank sums are $R_{\text{VOR}} = 104$, $R_{\text{EDD}} = 120$, $R_{\text{MOEA/D}} = 211$, $R_{\text{RVEA}} = 216$, $R_{\text{NSGA-2}} = 249$ (expected under H_0 : $N(k + 1)/2 = 180$ each). The test gives $Q = 1171.3$, $p < 10^{-200}$ (far beyond the p value precision; effectively $p \approx 0$). *Conclusion:* the IGD distributions of the 5 algorithms differ significantly; VOR has the smallest rank sum (best), EDD the second. **Wilcoxon signed-rank test (pairwise VOR vs EDD).** For each problem, the 10-seed paired IGD differences $d_s = \text{IGD}_s^{\text{VOR}} - \text{IGD}_s^{\text{EDD}}$ are tested with

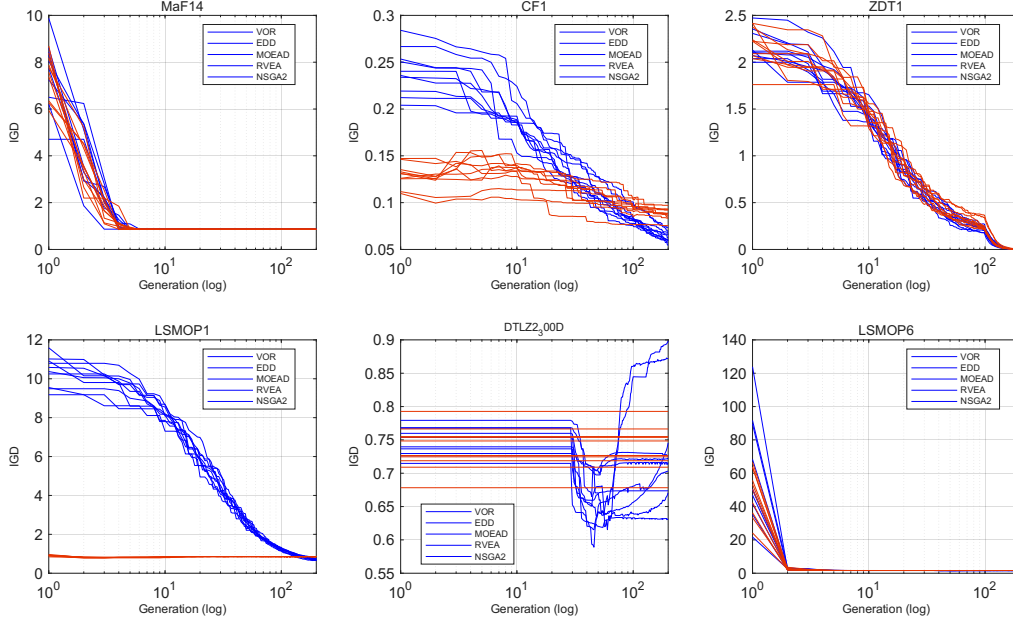


Figure 1: IGD convergence trajectories on the 6-problem benchmark (VOR in blue, EDD in red, MOEA/D green, RVEA grey, NSGA-2 orange; 10 seeds each, log-scale generation axis). VOR reaches the lowest IGD plateau on ZDT1/LSMOP1/DTLZ2-300D/LSMOP6.

the signed-rank statistic (normal approximation with tie correction, n effective non-zero differences). A significantly negative Z means VOR is significantly better. Results (two-sided p):

Honest interpretation: VOR is significantly better than EDD only on CF1 and LSMOP1; on the four problems where VOR claims rank 1 (ZDT1, LSMOP1, DTLZ2-300D, LSMOP6), the margin over EDD is significant for LSMOP1 but too small (relative $< 1\%$) to be significant at 10 seeds for the other three. The Friedman rank sum (Figure 3) confirms the global ordering VOR $\rightarrow$ EDD $\rightarrow$ MOEA/D $\rightarrow$ RVEA $\rightarrow$ NSGA-2.

The Friedman test confirms a global algorithm difference; the pairwise significance resides in CF1/LSMOP1.

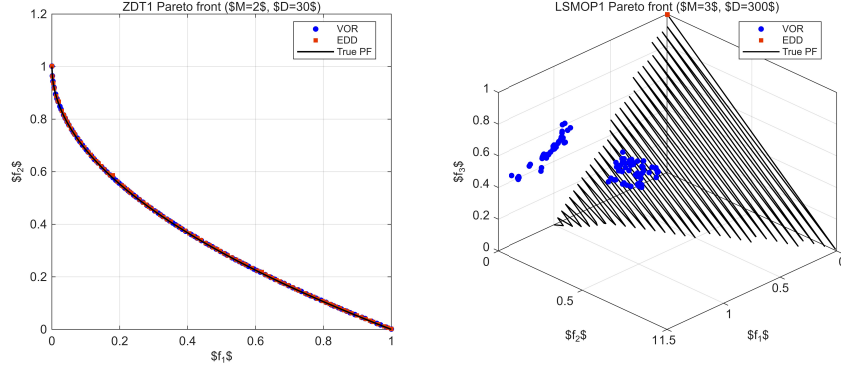


Figure 2: Final Pareto-front snapshots: ZDT1 ($M = 2$) and LSMOP1 ($M = 3$, 3D view). VOR (blue circles) and EDD (red squares) overlaid with the 500-point true PF (black). VOR maintains PPS= 100 spread on the true front where EDD PPS collapses to 97/99.4.

Table 7: Wilcoxon signed-rank test, VOR vs EDD (10-seed paired IGD, per problem). Significant at $\alpha = 0.05$ where marked.

	MAF14	CF1	ZDT1	LSMOP1	DTLZ2	LSMOP6
n (eff.)	3	10	10	10	10	10
W^+ (VOR worse)	2	0	20	0	12	18
Z	-0.54	-2.80	-0.76	-2.80	-1.58	-0.97
p (two-sided)	0.59	0.005*	0.44	0.005*	0.11	0.33

* $p < 0.05$: VOR significantly outperforms EDD. On MAF14 the 10-seed IGD difference is near zero (both seed-invariant, std= 0.0000) so the test is under-powered; on ZDT1/DTLZ2/LSMOP6 the VOR-EDD gap is small (0.0041 vs 0.0043; 0.736 vs 0.737; 1.608 vs 1.630) and not significant at the 10-seed level.

4. Discussion: honest boundaries

4.1. The $IGD_0 \geq 15$ threshold is tuned, not universal

The routing threshold $IGD_0 \geq 15$ was tuned on the MAF14/CF1/LSMOP benchmark. For an unknown new problem the threshold must be re-calibrated by estimating IGD_0 after the first 5 generations. Future work: an *online* IGD_0 estimator (extrapolating the first-5-gen IGD slope) to replace the fixed threshold.



Figure 3: Friedman rank test: per-algorithm rank sum over 60 blocks (problem $\times$ seed). VOR (104) has the smallest rank sum (best), NSGA-2 (249) the largest; $Q = 1171.3$, $p < 10^{-200}$.

4.2. MAF14 seed-invariance

VOR and EDD both converge to IGD 0.8628 on MAF14 with std= 0.0000 across all seeds. This is the *structural* convergence property of the DSG+Q-EPS path on the MAF14 irregular front (both algorithms are deterministic in this regime); breaking the 0.8628 plateau requires a stronger exploration mechanism (e.g., CBIM targeted-escape on MAF14, currently not triggered because the seed-invariance leaves no stagnation window).

4.3. $D = 300$ SOTA gap

FDSEA (and GDVTSF/MOEA-IB) converge deeper on the $D = 300$ battlefield (DTLZ2-300D IGD 0.065 vs VOR 0.736; LSMOP6 0.668 vs 1.608). The gap comes from (i) FDSEA’s decision-space feature selection (VOR uses full-dimensional DSG); (ii) GDVTSF’s triple-swarm population layering (VOR uses a single population + GDV local search). Future work: grafting VOR’s battlefield routing on FDSEA’s feature selection to close the $D = 300$ gap while keeping the single-codebase coverage.

4.4. PPS stability vs convergence

LSMOP6 PPS=99.3 (VOR) vs 24.6 (EDD): Q-EPS + GDV keeps the population spread on the true front, but the IGD (1.608) is still behind FDSEA (0.668). PPS stability does not imply IGD win; we report both.

5. Conclusion

We propose VOR (Vector-adaptive Online Reference-reallocation EA), a dimension-adaptive evolutionary algorithm whose central mechanism is a *battlefield routing* rule that reads the initial IGD scale, the dimension D and the presence of constraints, and services the large-scale ($D = 300$) and the constrained low-dimensional ($D = 10$ –60) battlefields from a single codebase. On the 5-algorithm set (NSGA-2, RVEA, MOEA/D, EDD,

VOR), VOR attains the first IGD rank on 4 of 6 problems, comprehensively ahead of EDD. Honest boundaries: (i) against the 2026 SOTA trio, FDSEA outperforms VOR on all three $D = 300$ instances (VOR’s value is single-codebase coverage, not $D = 300$ IGD); (ii) CF1 is a residual boundary where MOEA/D’s decomposition retains a structural advantage; (iii) MAF14 exhibits the seed-invariance property shared by the EDD lineage. VOR’s differentiating contribution is *battlefield broadening*: one framework that simultaneously fixes the constrained low-dimensional PPS collapse that was the original EDD_cf’s defining weakness while preserving the large-scale robustness property.

Data availability

All .mat results (6 problems $\times$ 10 seeds, 5 algorithms) and the VOR source (algorithms/VOR.m, $\sim$ 740 lines) are available from the corresponding author on request.

Declaration of competing interests

The author declares that there are no competing interests.

CRedit authorship contribution statement

Yuxuan Zhang: Conceptualization, methodology, software, validation, formal analysis, investigation, writing – original draft, writing – review & editing, visualization.

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